Civil Engineer to Data Scientist (with GenAI)

12-Month In-Depth Roadmap — 2026 Edition (updated & expanded from the original 8-month plan)

How to Reference the Learning Resources

All the core video resources used in this roadmap (Python, EDA, ML, Deep Learning, Cloud, Generative AI, FastAPI, RAG, Agentic AI, MCP and Claude Code) are the exact same ones from the companion document **“Resource 2026 — Complete Learning Roadmap”** we put together earlier — every link was checked and confirmed working. Rather than making you flip between two files, the direct link is repeated inline under each month's “Resources” section below. A consolidated reference table of every link, grouped by topic, is also included at the end of this document (see “Resource Reference Index”) so you always have a single place to look things up.

For topics that weren't part of that curated list (SQL, Power BI/Tableau, Git/GitHub, Docker/CI-CD) a suggested search term is given instead of a specific link — pick whichever well-rated free playlist you're comfortable with.

Roadmap at a Glance

Month	Focus
1	Math, Statistics & Python Foundations
2	Python for Data Analysis (Pandas/NumPy) + Git/GitHub
3	Exploratory Data Analysis & Visualization
4	SQL & BI Tools (Power BI / Tableau)
5	Machine Learning Foundations
6	Advanced ML, Feature Engineering & Model Evaluation
7	Deep Learning & Neural Networks
8	Cloud (GCP/AWS/Azure) + FastAPI Deployment
9	Generative AI, LLMs & LangChain
10	RAG Chatbots & Applied GenAI
11	Agentic AI (LangGraph), MCP & Claude Code
12	Capstone Projects, Portfolio & Job Readiness

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Month 1: Math, Statistics & Python Foundations

Focus: Build the programming and quantitative base everything else depends on.

Weekly Breakdown

- Week 1: Python syntax, variables, data types, loops, conditionals
- Week 2: Functions, OOP basics (classes/objects), error handling, file I/O
- Week 3: Statistics & probability — mean/median/mode, std dev, distributions, correlation; Excel basics
- Week 4: Practice problems (30+ small scripts) + mini project

Topics Covered

Core Python syntax and OOP, descriptive statistics, probability basics, Excel fundamentals (formulas, pivot tables).

Resources

- Python for Beginners (Full Course) — CodeWithHarry — https://youtube.com/playlist?list=PLu0W_9lII9agwh1XjRt242xIpHhPT2llg&si;=SyGwI9rmuLREWAWT
- 100 Days of Python Programming (Days 1–25) — CampusX — https://youtube.com/playlist?list=PLKnIA16_Rmvb1RYR-iTA_hzckhdONtSW4&si;=RNNMm0JhkQC6SF1V
- Optional: any beginner statistics playlist (search “statistics for data science”) for extra practice.

Mini / Milestone Project

Build a sales dataset dashboard in Excel, plus 3-4 small Python scripts (unit converter, expense tracker, basic data cleaner).

Suggested time commitment: *10–12 hrs/week*

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Month 2: Python for Data Analysis (Pandas/NumPy) + Git/GitHub

Focus: Move from plain Python to real data manipulation, and start version-controlling your work like a professional.

Weekly Breakdown

- Week 1: NumPy — arrays, vectorized operations, broadcasting
- Week 2: Pandas — Series/DataFrames, filtering, groupby, merging, handling missing data
- Week 3: Git & GitHub — commits, branches, pull requests; Jupyter/Google Colab workflow
- Week 4: Apply everything on a real dataset + push to GitHub

Topics Covered

NumPy arrays, Pandas data wrangling, missing-value handling, Git basics, notebook-based workflows.

Resources

- 100 Days of Python Programming (Days 26–60) — CampusX — https://youtube.com/playlist?list=PLKnIA16_Rmvb1RYR-iTA_hzckhdONtSW4&si;=RNNMm0JhkQC6SF1V
- Git & GitHub crash course (search “Git and GitHub for beginners” — any well-rated free playlist works).

Mini / Milestone Project

Clean and analyze an HR or Sales dataset in Pandas, write up 3-4 key insights, and publish the notebook on GitHub with a proper README.

Suggested time commitment: *10–12 hrs/week*

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Month 3: Exploratory Data Analysis & Visualization

Focus: Learn to actually understand data before modeling it — the skill most beginners skip and regret later.

Weekly Breakdown

- Week 1: Matplotlib & Seaborn — plots, styling, subplots
- Week 2: The EDA process — univariate/bivariate analysis, outliers, missing values
- Week 3: EDA + Feature Engineering applied to a real project
- Week 4: Full EDA report on a Kaggle dataset of your choice

Topics Covered

Data visualization, univariate/bivariate/multivariate analysis, outlier detection, feature engineering basics.

Resources

- Session 29: Exploratory Data Analysis | DSMP 2022-23 — CampusX — https://www.youtube.com/live/PPEHpg2RixQ?si=r9tITg14NYvKgihA
- Step by Step EDA & Feature Engineering in Data Science Projects — Krish Naik — https://youtu.be/xhB-dmKmzRk?si=-XfmsBOjTvkxHmTm

Mini / Milestone Project

Pick a Kaggle dataset (e.g. house prices, HR attrition) and produce a full EDA report with visuals and written insights — this becomes GitHub portfolio piece #1.

Suggested time commitment: *10 hrs/week*

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Month 4: SQL & BI Tools (Power BI / Tableau)

Focus: Learn to query data directly from databases and communicate insights through dashboards — core analyst/DS skills expected in almost every job.

Weekly Breakdown

- Week 1: SQL basics — SELECT, WHERE, GROUP BY, JOINS
- Week 2: Advanced SQL — subqueries, window functions, CTEs
- Week 3: Power BI or Tableau — connecting data, building visuals, KPIs
- Week 4: Build a full interactive dashboard

Topics Covered

SQL querying & joins, window functions, BI dashboard design, KPI definition.

Resources

- SQL for Data Science (search “SQL for data science full course” — CampusX SQL playlist recommended).
- Power BI or Tableau full course (search “Power BI full course” / “Tableau full course”).

Mini / Milestone Project

Build a Power BI or Tableau dashboard for HR or Sales analysis with at least 4 KPIs and 3 interactive visuals — portfolio piece #2.

Suggested time commitment: *8–10 hrs/week*

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Month 5: Machine Learning Foundations

Focus: Understand the core ML algorithms and how/why they work — not just how to call .fit().

Weekly Breakdown

- Week 1: Linear & Logistic Regression, cost functions, gradient descent
- Week 2: Classification — KNN, Decision Trees, Naive Bayes
- Week 3: Unsupervised learning — clustering (K-Means), PCA; evaluation metrics
- Week 4: End-to-end ML project

Topics Covered

Regression, classification, clustering, dimensionality reduction, train/test splits, evaluation metrics (accuracy, precision/recall, RMSE).

Resources

- 100 Days of Machine Learning — CampusX — https://youtube.com/playlist?list=PLKnIA16_Rmvbr7zKYQuBfsVkjoLcJgxHH&si;=AQ1td8bSZbnbYibt

Mini / Milestone Project

Build a customer churn prediction or house price prediction model, end-to-end from raw data to evaluated model — portfolio piece #3.

Suggested time commitment: *12–14 hrs/week*

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Month 6: Advanced ML, Feature Engineering & Model Evaluation

Focus: Go from a notebook model to something closer to production quality.

Weekly Breakdown

- Week 1: Feature engineering & feature selection techniques
- Week 2: Ensemble methods — Random Forest, Gradient Boosting, XGBoost
- Week 3: Hyperparameter tuning, cross-validation, avoiding overfitting
- Week 4: Structure an ML project properly (repo layout, config, tests) using Git

Topics Covered

Feature engineering, ensemble models, XGBoost, cross-validation, hyperparameter tuning, clean project structure.

Resources

- 100 Days of Machine Learning (continue deeper modules) — CampusX — https://youtube.com/playlist?list=PLKnIA16_Rmvbr7zKYQuBfsVkjoLcJgxHH&si;=AQ1td8bSZbnbYibt
- XGBoost official docs/tutorial (search “XGBoost tutorial for beginners”).

Mini / Milestone Project

Rebuild your Month 5 project with proper feature engineering, an XGBoost model, and cross-validated performance comparison — push to GitHub with a clean repo structure.

Suggested time commitment: *10–12 hrs/week*

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Month 7: Deep Learning & Neural Networks

Focus: Understand neural networks from first principles — this is what makes LLMs and GenAI make sense later instead of feeling like magic.

Weekly Breakdown

- Week 1: Perceptron, activation functions, forward/backward propagation
- Week 2: Convolutional Neural Networks (CNNs) for images
- Week 3: Recurrent Neural Networks (RNNs/LSTMs) for sequences
- Week 4: Project — train and evaluate a deep learning model

Topics Covered

Neural network fundamentals, CNNs, RNNs/LSTMs, TensorFlow/Keras or PyTorch basics.

Resources

- 100 Days of Deep Learning — CampusX — https://youtube.com/playlist?list=PLKnIA16_RmvYuZauWaPlRTC54KxSNLtNn&si;=TtWtWq-6UI2UCFij

Mini / Milestone Project

Build an image classifier using a CNN (e.g. cats vs dogs, or digit recognition) — portfolio piece #4.

Suggested time commitment: *12–14 hrs/week*

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Month 8: Cloud (GCP/AWS/Azure) + FastAPI Deployment

Focus: The month missing from the old roadmap. A model sitting in a notebook isn't a data science skill employers can use — you need to know how to serve it as an API and host it on the cloud.

Weekly Breakdown

- Week 1: Cloud fundamentals — compute, storage, IAM (skim all three: GCP, AWS, Azure to see the landscape)
- Week 2: Pick one cloud to go deep on (Azure recommended if you'll work with it professionally) + Docker basics
- Week 3: FastAPI — build a REST API around an ML model (request/response, Pydantic validation)
- Week 4: Deploy the FastAPI app to your chosen cloud platform

Topics Covered

Cloud fundamentals (compute/storage/IAM) across GCP/AWS/Azure, Docker basics, FastAPI, cloud deployment of a Python service.

Resources

- GCP Zero to Hero Series — Abhishek Veeramalla — https://youtu.be/N5rXROueKhw?si=Rpsj-WzRdSoiaiCm
- Free AWS Zero to Hero Course for DevOps Engineers — Abhishek Veeramalla — https://youtu.be/GkKNxyLp_V0?si=wIdYNI9GcfuKvm3a
- Azure Zero to Hero Course Syllabus (AZ-900/AZ-104) — Abhishek Veeramalla — https://youtu.be/10jm7Waan8M?si=MhmmTzCzcMnA_3h8
- FastAPI for Machine Learning — CampusX — https://youtube.com/playlist?list=PLKnIA16_RmvZ41tjbKB2ZnwchfniNsMuQ&si;=TYjhduFy1avM6v05

Mini / Milestone Project

Deploy your Month 6 classification model as a FastAPI service, containerize it with Docker, and host it live on GCP/AWS/Azure — portfolio piece #5 (a real, working, publicly accessible API).

Suggested time commitment: *12–14 hrs/week*

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Month 9: Generative AI, LLMs & LangChain

Focus: Shift from training models to building products on top of existing foundation models.

Weekly Breakdown

- Week 1: LLM fundamentals — tokens, context windows, how LLMs generate text
- Week 2: Prompt engineering — zero-shot/few-shot prompting, structured outputs
- Week 3: Embeddings & vector representations
- Week 4: LangChain framework — chains, memory, tool use

Topics Covered

LLM basics, prompt engineering, embeddings, the LangChain framework for building LLM applications.

Resources

- Generative AI using LangChain — CampusX — https://youtube.com/playlist?list=PLKnIA16_RmvaTbihpo4MtzVm4XOQa0ER0&si;=oAEAd_fii3ECytsQ

Mini / Milestone Project

Build a small LLM-powered app (a document summarizer or Q&A bot) using LangChain — portfolio piece #6.

Suggested time commitment: *10–12 hrs/week*

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Month 10: RAG Chatbots & Applied GenAI

Focus: Learn the single most in-demand GenAI pattern: connecting an LLM to your own data so it answers accurately instead of hallucinating.

Weekly Breakdown

- Week 1: RAG theory — retrieval + generation, chunking strategies
- Week 2: Vector databases — FAISS/Chroma/Pinecone, similarity search
- Week 3: Build a full RAG pipeline over your own PDFs/documents
- Week 4: Wrap the RAG pipeline in a FastAPI backend + a simple Streamlit/Gradio UI

Topics Covered

RAG architecture, chunking & embeddings, vector databases, wiring RAG into an API + UI.

Resources

- Complete RAG Playlist — Krish Naik — https://youtube.com/playlist?list=PLZoTAELRMXVM8Pf4U67L4UuDRgV4TNX9D&si;=SJDJMMT9_n7bGH1A
- FastAPI for Machine Learning (reference again for the backend) — CampusX — https://youtube.com/playlist?list=PLKnIA16_RmvZ41tjbKB2ZnwchfniNsMuQ&si;=TYjhduFy1avM6v05

Mini / Milestone Project

Build a RAG chatbot over your own documents (resume, notes, a company's public docs), served through FastAPI with a Streamlit/Gradio front end, deployed on cloud — portfolio piece #7 (flagship project).

Suggested time commitment: *12–14 hrs/week*

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Month 11: Agentic AI (LangGraph), MCP & Claude Code

Focus: The current frontier of applied AI: agents that plan, use tools, and complete multi-step tasks — plus the tooling standard (MCP) and AI coding agent (Claude Code) that make you dramatically faster at building all of this.

Weekly Breakdown

- Week 1: Agentic AI concepts — planning, memory, tool-calling; LangGraph basics
- Week 2: Build a multi-step agent workflow (e.g. research agent, support-ticket triage agent)
- Week 3: MCP (Model Context Protocol) — the standard way to connect agents to tools/data
- Week 4: Claude Code — use an AI coding agent to build/ship a real project faster

Topics Covered

LangGraph agent workflows, tool-calling, MCP fundamentals, agentic coding with Claude Code.

Resources

- Agentic AI using LangGraph — CampusX — https://youtube.com/playlist?list=PLKnIA16_RmvYsvB8qkUQuJmJNuiCUJFPL&si;=fL8A99RLb0n7FRE5
- Model Context Protocol — CampusX — https://youtube.com/playlist?list=PLKnIA16_Rmva_oZ9F4ayUu9qcWgF7Fyc0&si;=pzylTHxyGEh8OK29
- Claude Code Full Course 12 Hours: Build Real AI Projects (2026) — Mayank Aggarwal — https://youtu.be/05aY2LRIC3s?si=Wvkwkf-qwFaeaE2D

Mini / Milestone Project

Build an agent workflow automation project using LangGraph (with at least one MCP-connected tool), then use Claude Code to rapidly extend or refactor it — portfolio piece #8 (flagship project).

Suggested time commitment: *12–14 hrs/week*

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Month 12: Capstone Projects, Portfolio & Job Readiness

Focus: Consolidate everything into a portfolio that gets you interviews, and prepare to actually go through the interview process.

Weekly Breakdown

- Week 1: Pick 1-2 capstone projects that combine ML + GenAI + Cloud + FastAPI end-to-end
- Week 2: Polish all GitHub repos — clean READMEs, docs, basic tests, consistent structure
- Week 3: Build a portfolio website; write 3-4 LinkedIn posts documenting your projects
- Week 4: Resume, mock interviews (ML/DS + behavioral), start applying

Topics Covered

Project polish, documentation, personal branding (GitHub/LinkedIn/portfolio site), resume & interview preparation.

Resources

- Reuse and refine the projects from Months 5–11 — no new resources needed this month, this is execution & polish time.

Mini / Milestone Project

Ship 3–4 flagship, cloud-deployed projects: (1) end-to-end ML model deployment, (2) RAG chatbot, (3) agentic AI workflow, (4) optional — a dashboard/analytics project. Publish all on GitHub, link them from a portfolio website and LinkedIn.

Suggested time commitment: *10–12 hrs/week*

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Resource Reference Index

Every verified video resource used across this 12-month roadmap, in one place — the same set from the companion “Resource 2026” document, grouped by topic for quick lookup.

Topic	Resource	Link
Python	Python for Beginners (Full Course) — CodeWithHarry	youtube.com/playlist?list=PLu0W_9lII9agwh1XjRt242xIpHhPT2llg
Python	100 Days of Python Programming — CampusX	youtube.com/playlist?list=PLKnIA16_Rmvb1RYR-iTA_hzckhdONtSW4
EDA	Session 29: EDA | DSMP 2022-23 — CampusX	youtube.com/live/PPEHpg2RixQ
EDA	EDA & Feature Engineering — Krish Naik	youtu.be/xhB-dmKmzRk
Machine Learning	100 Days of Machine Learning — CampusX	youtube.com/playlist?list=PLKnIA16_Rmvbr7zKYQuBfsVkjoLcJgxHH
Deep Learning	100 Days of Deep Learning — CampusX	youtube.com/playlist?list=PLKnIA16_RmvYuZauWaPlRTC54KxSNLtNn
Cloud — GCP	GCP Zero to Hero Series — Abhishek Veeramalla	youtu.be/N5rXROueKhw
Cloud — AWS	AWS Zero to Hero (DevOps) — Abhishek Veeramalla	youtu.be/GkKNxyLp_V0
Cloud — Azure	Azure Zero to Hero (AZ-900/AZ-104) — Abhishek Veeramalla	youtu.be/10jm7Waan8M
Generative AI	Generative AI using LangChain — CampusX	youtube.com/playlist?list=PLKnIA16_RmvaTbihpo4MtzVm4XOQa0ER0
FastAPI	FastAPI for Machine Learning — CampusX	youtube.com/playlist?list=PLKnIA16_RmvZ41tjbKB2ZnwchfniNsMuQ
RAG	Complete RAG Playlist — Krish Naik	youtube.com/playlist?list=PLZoTAELRMXVM8Pf4U67L4UuDRgV4TNX9D
Agentic AI	Agentic AI using LangGraph — CampusX	youtube.com/playlist?list=PLKnIA16_RmvYsvB8qkUQuJmJNuiCUJFPL
MCP	Model Context Protocol — CampusX	youtube.com/playlist?list=PLKnIA16_Rmva_oZ9F4ayUu9qcWgF7Fyc0
Bonus	Claude Code Full Course 12 Hrs — Mayank Aggarwal	youtu.be/05aY2LRIC3s

For full descriptions, “why it matters” context, and study order, see the companion document “Resource 2026 — Complete Learning Roadmap.” All links above were verified working when this roadmap was prepared.

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Final Summary & Advice

In 12 months, you go from Python basics to a working, cloud-deployed portfolio spanning Data Analysis, Machine Learning, Deep Learning, Generative AI and Agentic AI — using the same trusted CampusX / Krish Naik / Abhishek Veeramalla resources, now including the Cloud and Agentic AI/MCP/Claude Code layers the old 8-month plan didn't cover.

Final Advice

- Consistency beats intensity — a realistic 10–14 hrs/week for 12 months beats burning out in 2 months.
- Every month ends with a project. Do not skip these — they are what becomes your portfolio.
- Certificates matter far less than a GitHub portfolio with real, working, deployed projects.
- Cloud + FastAPI (Month 8) is non-negotiable — it's what separates “can build a model” from “can ship a product.”
- Keep learning, sharing progress, and building publicly on LinkedIn throughout — not just in Month 12.

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